

NATIONAL JUNIOR COLLEGE
SENIOR HIGH 2 Preliminary Examination
Higher 3

MATHEMATICS

9824

18 September 2015

3 hours

Additional Materials: Answer Paper
 List of Formulae (MF15)
 Cover Sheet

READ THESE INSTRUCTIONS FIRST

Write your name, registration number, subject tutorial group, on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for diagrams or graphs.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of **6** printed pages.



National Junior College

Answer **all** the questions.

- 1 (i) Use the formula for the sum of an infinite geometric series to prove that

$$a_n = \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \dots$$

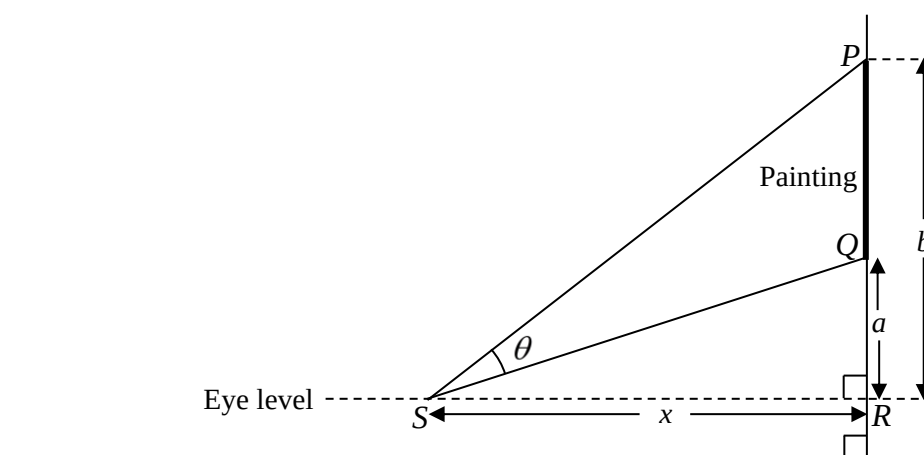
is less than $\frac{1}{n}$ for each positive integer n . [2]

- (ii) By considering the Maclaurin series for e^x , verify that $e = \lfloor e \rfloor + a_1$ and $2e = \lfloor 2e \rfloor + a_2$, where $\lfloor x \rfloor$ is the integer part of x . [2]

- (iii) Show that $a_n = n!e - \lfloor n!e \rfloor$. [4]

- (iv) Hence show that e is irrational. [4]

- 2 A tourist at S looks at a painting, PQ , on the wall as shown in the diagram below. The vertical distance between the eye level of the tourist, and the bottom and top of the painting respectively are a units and b units respectively. The horizontal distance between the tourist and the painting is denoted by x . The viewing angle of the tourist, $\angle PSQ$, is denoted by θ .



- (i) Show, by differentiation, that the maximum value of θ occurs at $x = \sqrt{ab}$. [7]

- (ii) Explain why the maximum value of $\tan \theta$ also occurs at $x = \sqrt{ab}$ and find the maximum value of $\tan \theta$ in terms of a and b . (Your final answer should not contain any inverse trigonometric functions.) [4]

- (iii) Show that, when θ achieves its maximum value, SR will be tangent to the circle passing through the points P , Q and S . [3]

- 3 A continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ is said to be **convex** on $\mathbb{R}$ if, for all $a, b \in \mathbb{R}$ and for all $\lambda \in [0, 1]$,

$$f(\lambda a + (1 - \lambda)b) \leq \lambda f(a) + (1 - \lambda)f(b). \quad (*)$$

- (i) Show that the sum of two convex functions is a convex function. [2]
- (ii) Show that the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = kx^2$, where $k > 0$, is convex on $\mathbb{R}$. [4]
- (iii) Prove or disprove the following statements:
 - (a) Linear functions are convex.
 - (b) The product of two convex functions is convex. [5]
- (iv) If the inequality sign in (*) is replaced with “<”, the function is said to be **strictly convex** on $\mathbb{R}$. Prove, by contradiction or otherwise, that a function which is strictly convex on $\mathbb{R}$ achieves its minimum value at most once. [3]

- 4 (a) Let T_n denote the number of integer triples (a, b, c) such that $|a| + |b| + |c| = n$, where n is a positive integer.

- (i) State the value of T_1 . [1]
- (ii) Find T_n in terms of n . Simplify your answer. [5]

- (b) Let $S(n, k)$ denote the number of ways to divide $\{1, 2, 3, \dots, n\}$ into k disjoint non-empty subsets, where $n \geq 2$.

- (i) Show $S(n, n-1) = \frac{n(n-1)}{2}$. [2]
- (ii) Find $S(n, n-2)$ in a similar form. [3]

- 5 (a) How many positive integers are divisors of at least one of the numbers 12^{50} , 45^{20} and 50^{15} ? [5]

- (b) Let a_n denote the number of ways that a $3 \times n$ checkerboard can be filled by n non-overlapping 1×3 dominos.

- (i) Write down the values of a_1 , a_2 and a_3 . [1]

- (ii) Find a recurrence relation for a_n where $n \geq 4$, and use it to find the value of a_9 . [3]

- (iii) Show that $a_{2n} > a_n^2$ for $n \geq 2$. [2]

Show that, when a 9×15 checkerboard is completely filled by non-overlapping 1×3 dominos, at least one line between two adjacent rows or two adjacent columns will be cut by at least 5 dominos. [3]

- 6 The motion of a projectile can be modelled by the following differential equation relating the speed, s , of the projectile with the angle of inclination, θ ,

$$\frac{ds}{d\theta} - s \tan \theta = as^3 \sec \theta,$$

where a is a constant.

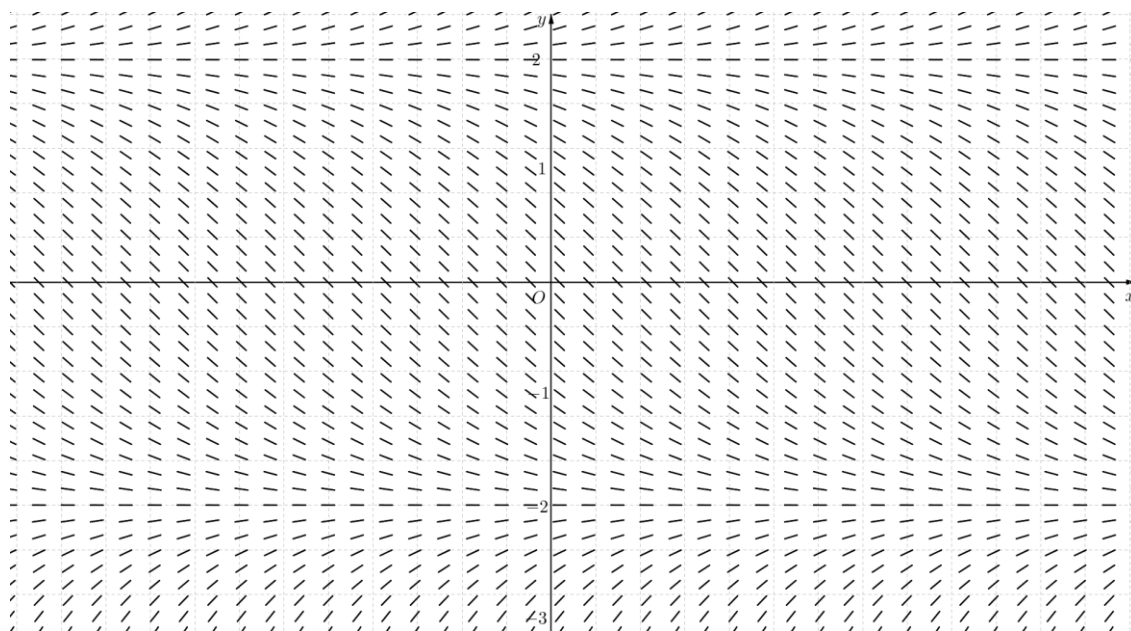
- (i) Using the substitution $u = s^{-2}$, show that

$$s = \frac{\sec \theta}{\sqrt{a[m - \sec \theta \tan \theta - \ln(\sec \theta + \tan \theta)]}},$$

for some constant m . [8]

- (ii) Given that θ is sufficiently small for terms in θ^2 and higher powers to be ignored, show that the speed of the projectile can be approximately expressed as a linear function of θ . [3]

7 (a)



The sketch above shows the slope field diagram for a differential equation of the form

$$\frac{dy}{dx} = f(y).$$

Sketch a phase line diagram for this differential equation and comment on the stability of the equilibrium values. [4]

- (b) The grass patch next to Bukit Mahti MRT Station has been infested with rats in recent weeks. The population size, x , in t days (since the infestation was discovered) can be modelled by the differential equation

$$\frac{dx}{dt} = 0.1x - 0.0002x^2.$$

To solve the infestation problem, the National Environment Authority has employed a group of pest controllers to exterminate the rodents. The pest controllers manage to kill a constant number of rats, h , per day since the infestation was discovered.

- (i) Sketch a bifurcation diagram for h , including the locus of equilibrium points. Label clearly the coordinates of the point(s) of intersection between this locus and the vertical axis, as well as the bifurcation value for h . [4]

Assume now that $h = 12$ and that there were initially x_0 rats.

- (ii) It is given that despite the pest controllers' efforts, the rat population size steadily increases and eventually stabilises at a fixed number. Use your sketch in part (b)(i) to find the range of values of x_0 . [2]

- (iii) If given instead that $x_0 = 250$, find x in terms of t . [5]

- 8** In a spring-mass-dashpot system, a body with mass $m = 0.5$ kg is attached to the end of an elastic spring as well as to a dashpot, and is initially released from rest at a displacement of 0.2 m from its equilibrium position O .

- (i)** Briefly explain why the motion of the body can be modelled by the differential equation

$$\ddot{x} + a\dot{x} + bx = 0,$$

for some positive constants a and b , where x m is the displacement of the body from O at time t s. [2]

- (ii)** Given that $b = 4$, find the set of values of the damping constant for which the motion would be over-damped. [2]

- (iii)** Suppose now that $a = 3$ and $b = 8.5$. Find x in terms of t and sketch a displacement-time graph that corresponds to the first 3 seconds of the motion of the body. [5]

-- END OF PAPER --